Daviti Gochitashvili

+1 (607) 232-2459 | Vestal, NY | datogochitashvili@gmail.com | linkedin.com/in/davitgochitashvili | GitHub

EDUCATION

Binghamton University — *Vestal, NY*
PhD in Computational Condensed Matter Physics — Aug. 2020 – Dec. 2025 (Expected)

Free University of Tbilisi — *Tbilisi, Georgia*
BS in Physics — Sep. 2015 – Aug. 2019

FINANCIAL INDUSTRY EXPERIENCE

Bank of Georgia, Analyst – Developer, *Tbilisi, Georgia* — Jul. 2019 – Aug. 2020

- Developed API web services for the Core Banking System using Java and PL/SQL, enhancing system responsiveness, streamlining internal workflows, and enabling efficient handling of high-volume banking transactions.
- Designed and implemented quantitative finance and banking-related functions for the Cash Operations and Liens Agile (Scrum) development squad, enabling data-driven decision-making and faster project delivery.
- Collaborated with Revenue Service officials to gather requirements, analyze complex banking rules, ensure compliance, and convert them into actionable system functions and automated solutions that improved efficiency.
- Accelerated cash counting operations by ~ 5 times and automated money counting machines to generate detailed operational and regulatory reports, leveraging complex SQL queries for analytics and process validation.

ACADEMIC EXPERIENCE

Binghamton University, Graduate Researcher/Associate — Sep. 2020 – Present

4 peer-reviewed publications (20 citations); see Google Scholar profile.

- Contributed to development of the open-source MAISE package (C and Python), streamlining construction, testing, and application of accurate ML models of atomic interactions, with integrated Slurm/HPC workflows.
- Utilized the accurate neural network interatomic potentials for discovering new stable compounds using evolutionary algorithm. Screened configurational space, refining the thermodynamic convex hull with 29 materials.
- Implemented hyperdimensional optimization enhancing the efficiency to navigate complex energy surfaces; Large-scale statistical benchmarkings confirmed faster sampling and improved global-minimization attainment.

Binghamton University, Teaching Assistant — Sep. 2020 – Present

- Led two discussion sections of 30 students each for general physics courses. Provided individual student support.
- Conducted lab demonstrations, set up equipment for experiments, assisted students during lab course time.

PERSONAL PROJECTS

- Implemented option-pricing models (Black–Scholes, binomial trees, Monte Carlo) and numerical PDE solvers, with simulation frameworks to compute Greeks and stress-test strategies under varying volatility. (Repo link)
- Backtested a free-float–weighted equity index to analyze correlations and portfolio diversification. (Repo link)

SKILLS

Technical skills:	Machine learning, data analysis & visualization, statistics, stochastic calculus.
Programming languages:	Python (NumPy, Pandas, SciPy, matplotlib, scikit-learn), PL/SQL, Java, C++.
Software:	Git, Bash, Linux, Slurm, MATLAB, Wolfram Mathematica, Microsoft Excel.
Languages:	Georgian (native), English (fluent), Russian (conversational), German (basic).

HONORS AND FELLOWSHIPS

Provost's Doctoral Summer Fellowship (Binghamton University). — 2020 – 2024
Knowledge Fund Scholarship (Free University of Tbilisi). — 2015 – 2019